AUTOMATED PLANETARY PERCEPTION SYSTEM

Part 14: Multi-Modal Active Perception & Degradations

Level 3, Volume 2 of 5: Stereo Navcam, 3D Flash LiDAR, IMU Stochastic Errors, Solar Glare & Negative Obstacles

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VOLUME SUMMARY: *Specifies multi-sensor active perception payloads (Stereo Navcam, 3D Flash LiDAR, Tactical IMU), atmospheric dust extinction (optical depth tau), directional solar glare blooming, negative obstacle detection, and calibrated 3D epistemic voxelization.*

Chapter 240. Executive Summary & Active Perception Scope

Context & Operational Boundaries: Defines the Level 3 multi-sensor active perception suite and environmental degradation test matrix.

• **Multi-Modal Payload:** Integrates stereo high-resolution Navcams, 3D Flash LiDAR / ToF scanners, and tactical 6-axis IMUs.

• **Real-World Degradations:** Explicitly simulates atmospheric dust storms, direct solar glare flare, thermal optical distortions, and negative drop-off obstacles.

• **Information Boundary:** Perception receives exclusively degraded raw sensor streams; perfect ground-truth voxel grids remain hidden.

ty	Operational Range	Primary Measurement	Simulated Degradation Mode
	0.5 m to 30.0 m	RGB Texture + Disparity Depth	Solar Glare, Dust Extinction, Thermal Bias
	0.2 m to 15.0 m	Direct Time-of-Flight Point Clouds	Incidence Angle Drop-off, Range Noise
	Continuous (200 Hz)	3-Axis Specific Force & Angular Rate	Allan Variance Gyro Drift, Temp Bias
	Continuous (50 Hz)	Wheel Angular Velocity	Slippage Discrepancy vs Visual Odometry

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Chapter 240. Chapter 240.1: Active Perception Specification 1

Level 3 multi-modal perception and environmental robustness specification page 4.

Equation (244.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 4.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 240. Chapter 240.2: Active Perception Specification 2

Level 3 multi-modal perception and environmental robustness specification page 5.

Equation (245.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 5.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 240. Chapter 240.3: Active Perception Specification 3

Level 3 multi-modal perception and environmental robustness specification page 6.

Equation (246.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 6.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 241. Chapter 241.1: Active Perception Specification 4

Level 3 multi-modal perception and environmental robustness specification page 7.

Equation (247.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 7.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 241. Chapter 241.2: Active Perception Specification 5

Level 3 multi-modal perception and environmental robustness specification page 8.

Equation (248.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 8.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 241. Chapter 241.3: Active Perception Specification 6

Level 3 multi-modal perception and environmental robustness specification page 9.

Equation (249.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 9.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 242. Chapter 242.1: Active Perception Specification 7

Level 3 multi-modal perception and environmental robustness specification page 10.

Equation (250.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 10.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 242. Chapter 242.2: Active Perception Specification 8

Level 3 multi-modal perception and environmental robustness specification page 11.

Equation (251.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 11.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 242. Chapter 242.3: Active Perception Specification 9

Level 3 multi-modal perception and environmental robustness specification page 12.

Equation (252.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 12.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 243. Chapter 243.1: Active Perception Specification 10

Level 3 multi-modal perception and environmental robustness specification page 13.

Equation (253.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 13.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 243. Chapter 243.2: Active Perception Specification 11

Level 3 multi-modal perception and environmental robustness specification page 14.

Equation (254.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 14.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 243. Chapter 243.3: Active Perception Specification 12

Level 3 multi-modal perception and environmental robustness specification page 15.

Equation (255.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 15.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 244. Chapter 244.1: Active Perception Specification 13

Level 3 multi-modal perception and environmental robustness specification page 16.

Equation (256.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 16.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 244. Chapter 244.2: Active Perception Specification 14

Level 3 multi-modal perception and environmental robustness specification page 17.

Equation (257.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 17.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 244. Chapter 244.3: Active Perception Specification 15

Level 3 multi-modal perception and environmental robustness specification page 18.

Equation (258.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 18.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 245. Chapter 245.1: Active Perception Specification 16

Level 3 multi-modal perception and environmental robustness specification page 19.

Equation (259.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 19.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 245. Chapter 245.2: Active Perception Specification 17

Level 3 multi-modal perception and environmental robustness specification page 20.

Equation (260.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 20.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 245. Chapter 245.3: Active Perception Specification 18

Level 3 multi-modal perception and environmental robustness specification page 21.

Equation (261.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 21.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 246. Chapter 246.1: Active Perception Specification 19

Level 3 multi-modal perception and environmental robustness specification page 22.

Equation (262.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 22.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 246. Chapter 246.2: Active Perception Specification 20

Level 3 multi-modal perception and environmental robustness specification page 23.

Equation (263.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 23.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 246. Chapter 246.3: Active Perception Specification 21

Level 3 multi-modal perception and environmental robustness specification page 24.

Equation (264.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 24.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

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Equation (265.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 25.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 247. Chapter 247.2: Active Perception Specification 23

Level 3 multi-modal perception and environmental robustness specification page 26.

Equation (266.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 26.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 247. Chapter 247.3: Active Perception Specification 24

Level 3 multi-modal perception and environmental robustness specification page 27.

Equation (267.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 27.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 248. Chapter 248.1: Active Perception Specification 25

Level 3 multi-modal perception and environmental robustness specification page 28.

Equation (268.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 28.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 248. Chapter 248.2: Active Perception Specification 26

Level 3 multi-modal perception and environmental robustness specification page 29.

Equation (269.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 29.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 248. Chapter 248.3: Active Perception Specification 27

Level 3 multi-modal perception and environmental robustness specification page 30.

Equation (270.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 30.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 249. Chapter 249.1: Active Perception Specification 28

Level 3 multi-modal perception and environmental robustness specification page 31.

Equation (271.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 31.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 249. Chapter 249.2: Active Perception Specification 29

Level 3 multi-modal perception and environmental robustness specification page 32.

Equation (272.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 32.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 249. Chapter 249.3: Active Perception Specification 30

Level 3 multi-modal perception and environmental robustness specification page 33.

Equation (273.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 33.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 250. Chapter 250.1: Active Perception Specification 31

Level 3 multi-modal perception and environmental robustness specification page 34.

Equation (274.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 34.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 250. Chapter 250.2: Active Perception Specification 32

Level 3 multi-modal perception and environmental robustness specification page 35.

Equation (275.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 35.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 250. Chapter 250.3: Active Perception Specification 33

Level 3 multi-modal perception and environmental robustness specification page 36.

Equation (276.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 36.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)

Chapter 251. Chapter 251.1: Active Perception Specification 34

Level 3 multi-modal perception and environmental robustness specification page 37.

Equation (277.1): Radiometric Sensor & Uncertainty Model

```
I_obs(u, v) = I_true(u, v) * exp( -tau_dust * d_ray ) + I_glare(u, v) + N( 0, sigma_cam^2 ) u(x, y, z) = 1.0 - exp( - ( sigma_lidar^2 + sigma_stereo^2 ) / K_epistemic )
```

Where: tau_dust is Martian optical depth, d_ray is sightline distance, I_glare is veiling glare flux, u is epistemic voxel uncertainty in [0, 1].

- **Degradation Sentinel:** Monitors signal-to-noise ratio and flags degraded perception regimes across operational sector 37.
- **Epistemic Voxelization:** Unknown, shadow-occluded, and low-confidence ground cells are assigned u = 1.0 (blocked by default).

Parameter	Nominal Value	Degraded Storm Limit
Dust Optical Depth tau	0.45	3.80 (Severe Dust Storm)
Solar Flare Veiling	0.05	0.85 (Direct Sun In-Frame)
LiDAR Return Ratio	98.5%	42.0% (High Incidence Angle)